MSG No. 63.463 $Cmc'm'$ [Type III, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[13,14,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,3,4]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[13,14,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `m2'm'`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,4,5,8]`
2	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[2,3,6,7]`
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[9,12,13,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[10,11,14,15]`

Table 4: Wyckoff site: `8d`, site symmetry: `-1`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$	`[1,11]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$	`[2,12]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ 0\right]$	`[3,9]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$	`[4,10]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[5,15]`
6	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[6,16]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[7,13]`
8	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[8,14]`

Table 5: Wyckoff site: `8e`, site symmetry: `2..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,2]`
2	$[-x,\ 0,\ 0]$	`[3,4]`
3	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[5,6]`
4	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[7,8]`

continued ...

Table 5

No.	position	mapping
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[11,12]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[13,14]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[15,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `m..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,4]`
2	$[0,\ -y,\ -z]$	`[2,3]`
3	$\left[0,\ y,\ \frac{1}{2}-z\right]$	`[5,8]`
4	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	`[6,7]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[9,12]`
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[10,11]`
7	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,16]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[14,15]`

Table 7: Wyckoff site: `8g`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,8]`
2	$\left[x,\ -y,\ \frac{3}{4}\right]$	`[2,7]`
3	$\left[-x,\ -y,\ \frac{3}{4}\right]$	`[3,6]`
4	$\left[-x,\ y,\ \frac{1}{4}\right]$	`[4,5]`
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[9,16]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[10,15]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[11,14]`
8	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[12,13]`

Table 8: Wyckoff site: `16h`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x,\ -y,\ -z]$	`[2]`
3	$[-x,\ -y,\ -z]$	`[3]`
4	$[-x,\ y,\ z]$	`[4]`
5	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[5]`
6	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[6]`
7	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[x,\ y,\ \frac{1}{2}-z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[9]`
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[10]`
11	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	`[11]`

continued ...

Table 8

No.	position	mapping
12	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[12]
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[15]
16	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[16]